2 Revealing Hegemony: Agonistic Information Design

Money and politics have always gone together, and wealth has wielded influence since the beginnings of democracy. So it is not surprising that elected representatives are influenced by the individuals, corporations, and interest groups that fund their campaigns. With improved access to data and new ways of expressing information, novel computational forms illuminate in greater detail and with new cleverness this age-old entanglement of money and politics. For instance, the computational visualization *State-Machine: Agency* (Carlson and Cerveny 2005)[1] depicts the relationship between United States senators and their campaign contributors (figure 2.1). In the digital project, senators are represented as either red or blue circles, depending on their political party affiliation, and the size of their circle is determined by the total amount of campaign funds received. The circles are placed on the screen relative to the amount of funding received from one of three variable funding sources. Each funding source is represented by a plus sign, whose size is determined by the total number of dollars contributed to all campaigns. Funding sources such as lawyers and law firms, which contributed $744,660,550 to all Senate campaigns in 2007, are visually represented as proportionately larger than funding sources such as public-sector unions, whose contributions totaled $7,935,381. A senator who received more funding from lawyers and law firms than from public-sector unions would thus appear closer to that representative plus sign. Using menus in the interface, users can select different combinations of funding sources, resulting in the display of different images. Each new image reveals a different pattern of associations between senators and campaign contributions.

In many ways, *State-Machine: Agency* is familiar as a computational visualization. It draws together and renders a complex set of data; it makes use of standard visual cues such as shape, size, and color to assign

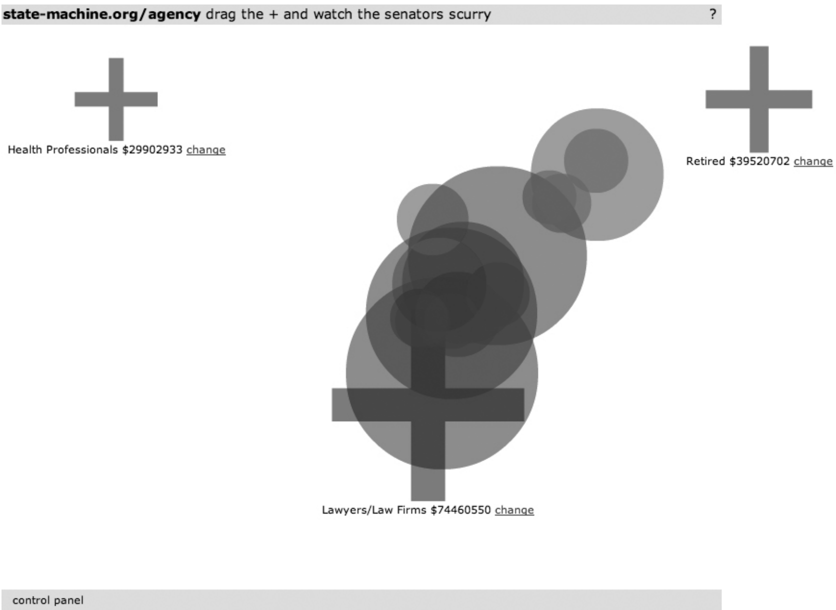

Figure 2.1
Max Carlson and Ben Cerveny, *State-Machine: Agency* (2005), http://state-machine.org

significance to the data; and through a simple interface, it enables user interaction (by manipulating variables, users can explore the data and generate new representations based on their interests and desires). But beyond these familiar qualities and mechanisms, *State-Machine: Agency* is distinctive in ways that make it exemplary of adversarial design. Foremost, the visualization assumes a decidedly political stance. Because of this explicit political stance, it diverges sharply from the tradition of visualization as a scientific technique that is presumed devoid of bias. This political stance can be immediately perceived in the title. In computer science, a state machine is a model of a set of possible behavioral relations between input, output, and action in which the status of the system is known, stored, and available to be operated on by procedural means. The title thus draws an association between an algorithmic process, which suggests a notion of the determined or expected, and the ways in which a particular political state operates. In case the perspective is not clear from the title of the project, the introductory screen to the visualization states, "Money drives the American political system."

The political stance is not expressed only by discursive means. The political stance is an integral part of the visualization itself, expressed through the interactive qualities of the visualization. Clicking and dragging on a representative circle allows the user to pull a senator away from a funding source momentarily, but as soon as the cursor is released, the circle bounces back into place, its position defined by its relation to the selected funding sources and their comparative contributions. The design of the visualization does more than simply present data. It expressively renders the associations between data, illustrating in an interactive form the notion that politicians are bound to their positions, which are defined by those that give them money.

Visualizations such as *State-Machine: Agency* are a distinct kind of computational object that is emblematic of computational information design. They merit attention because they have become one of the most recognized forms of expressive computational media and constitute an area of considerable design activity. Far extending the purview of their origins in the sciences, computational visualizations have become a familiar cultural form. They commonly appear as a means of explanation in popular visual media, such as print advertising and television news programs. Within some Web sites, computational visualizations have transitioned from being explanatory support material to being the content itself. The *New York Times*, for example, has developed a series of computational visualizations that are self-contained "news stories" of a new kind. The *Naming Names* (Corum and Hossain 2007)[2] visualization is one such example. The visualization allows users to explore who referred to whom in the Democratic and Republican debates in the early part of the 2008 United States presidential election. By interacting with the visualizations, users can discover patterns of referencing among candidates over time. The inclusion of the quotes in which a candidate was named further allows users to develop an understanding of the context of the references. Through this combination of the techniques of information design and the capacities of computation, *Naming Names* intimates a notion of visualization as a kind of journalism.[3]

In addition to their use in news media, computational visualizations are also embedded within other media forms as aesthetic elements or as tools for learning, play, or reflection. Contemporary films, particularly those involving science fiction, regularly use computational visualizations and other forms of information design as visual props to contribute to a contemporary technological aesthetic. As design critic Peter Hall (2008, 122) notes, "Cascading veils of information, as famously depicted in the 1999 film *The Matrix*, have become a defining signifier of our age." Visualization and the aesthetics and practices of computational information design express a cultural moment in which information is abundant, and designers are challenged to make sense of and manipulate that information for social or cultural effect.

Because computational visualization and computational information design are common cultural forms, designers and artists are experimenting to extend the contexts, content, and purposes of information design in new directions. Consider *The Dumpster* (Levin, Nigam, and Feinberg 2006),[4] which depicts breakups mined from an online journal service, allowing users to surf these moments, or consider *We Feel Fine* (Harris and Kamvar 2005),[5] which depicts emotional states gathered from blog postings and allows users to sort and combine feelings and demographics into ever-changing representations of mood. These visualizations are noteworthy for their visual and interactive inventiveness. They elide distinctions between art and design as artists engage the practice of information design and designers veer from rote communication toward authorial expression. They also broaden the scope of information design beyond representing the objective and factual to attempt to represent the subjective and affective. In doing so, such experiments by designers and artists simultaneously utilize and interpret the forms and processes of information design toward new ends.

Some of these experiments by artists and designers are political in an unequivocal sense: they expose and document power structures and networks of influence. This is evident in *State-Machine: Agency,* which depicts a set of financial forces that are exerted by special-interest groups through campaign funding. In such works, artists and designers employ the principal qualities of computation toward decidedly agonistic ends. Those principal qualities and the tactics by which they represent and perform political relations are the subject of this chapter.

Computational Information Design

Information design is the practice of giving form to data so that the data become meaningful. In this context, data are a raw material. The data can be of many types, from digits streaming across the Internet to electrical stimulation in the brain. But such data has no, or minimal, meaning associated with it. In contrast, information is data that have been given structure and shape, translated and contextualized in a way that allows them to be used as the basis for understanding or action. The practices of information design usually center on activities of rendering data, encompassing the dual meaning of rendering as both an activity of processing and an activity of producing an image or representation.

Like design generally, information design extends beyond any single discipline. It draws from multiple fields, including graphic design, writing and technical communication, information science, cognitive science, and computer science. The products of information design are equally diverse, including typography, layout, text, diagrams, illustrations, documentary photography, maps, and visualizations. Information design as a practice reflects the constitution and role of information generally within society: the practices and forms of information design respond to the changing qualities of data. As the mediums of data transmission and consumption and the materiality of data have changed in the late twentieth and early twenty-first century, so too has the practice of information design changed. Computation has affected the ways in which data are processed and formed into representations and the qualities of those representations. Understanding the practice of computational information design and providing insights into what it means to do design with computation requires identifying and understanding the principal qualities of computation. These qualities shape the practice of information design and provide particular affordances for agonistic expression.

Computational Media and Information Design

In *Visualizing Data*, designer Ben Fry (2008) provides a process for doing computational information design that is structured through seven stages—acquire, parse, filter, mine, represent, refine, and interact. This process is robust enough to guide a practitioner through many computational information design problems. It provides a flexible operational framework for working with data that is informed by Fry's extensive experience in using computation as a medium for information design. But for both the practicing designer and the design scholar, such processes need to be complemented with an understanding of the qualities of computation that underlie such an operational framework and that make computation a distinctive medium for design.

For this discussion of computation and information design, I draw on digital media studies, where much of the richest discussion of computation as a medium has taken place. In part, this may be due to the legacy of media studies and its concern with the characteristics of a given medium, whether radio, film, video, or digital. Extending this concern to the work of identifying the qualities of computation, there is value in understanding what distinguishes a medium, while avoiding essentializing the medium.

Drawing from digital media studies, the three principal qualities of computation that characterize computational information design are procedurality, transcoding, and the network as a medium of storage, access, and exchange. In computational information design, these qualities combine to render data into information in new ways and to produce new forms of expression. The network as a medium of storage, access, and exchange enables the culling and referencing of an enormous diversity of data; transcoding enables those data to be converted into and across a variety of formats; and procedurality enables the authoring of code to perform these conversions and structure representations algorithmically.

The term *procedurality* refers to the operational characteristic of computation: computation works by executing a set of rules for symbol manipulation. These rules most commonly come in the form of software or, more colloquially, code, which formalizes relationships between symbols and determines how the rules are to be executed, thereby defining the capacities and form of a given computational expression. For many digital media scholars, procedurality is the foundation of computational media. For Janet Murray (1997, 71), procedurality is one of the four essential properties of the digital environment, and she characterizes it as "the defining ability [of the computer] to execute a series of rules." For Ian Bogost (2007), procedurality is central to computation and the basis of a new form of rhetoric performed with computational media, which he calls procedural rhetoric. Both Murray and Bogost identify procedurality as the prime factor in the expressive capabilities of computational media. As Bogost (2007, 5) states, "Computation is representation, and procedurality in the computational sense is a means to produce that expression."

Although Lev Manovich does not use the term *procedurality*, his notion of the "new media object" references procedurality. Manovich (2001, 47) describes the new media object as being "subject to algorithmic manipulation" via programming and then goes on to claim programmability as "the most fundamental quality of new media that has no historical precedent." In effect, Manovich, echoes the centrality of procedurality to computational media. Manovich (2001, 47) further develops the notion of the new media object through a set of general principles: numerical representation, modularity, variability, automation, and transcoding. Of these, transcoding—the capacity of a new media object to be converted from one format to another—is particularly salient to practices and products in computational information design. As an example, consider any number of interactive map products or location-based services that integrate spatial data, photographs, and user-generated content.[6] These products and services are made possible by the shared structure of the code and the subsequent capacity to weave together digital content, dynamically integrating strings of text with the vector graphics of maps along with animated bitmaps of graphs and images. Such transcoding depends on and expresses Manovich's other four principles. Because of numerical representation and modularity, transcoding can occur, and through transcoding the variability of computational objects is expressed, increasingly in an automated or semiautomated manner.

Finally, the practice of information design today is, in large part, fashioned by the growth of the network as a medium of storage, access, and exchange. The Internet as both a repository of digital data and a medium for the transmission of data has significantly influenced the practice of computational information design by providing access to a massive amount and diversity of data. Photo-sharing services, text chats, and social network sites are all used as sources of data. Reports on stock market activity, environmental conditions, and the weather are also available in real time or near real time. Many databases produced by government agencies and nongovernmental organizations are readily available, covering the broadest of perspectives, from the Central Intelligence Agency's *World Factbook*[7] database of countries and territories to the *Greenpeace Blacklist*[8] database, which registers fishing vessels and companies engaged in illegal, unregulated, and unreported fishing. In addition, commercial databases are readily obtainable for purchase, with cost, not content, being the constraining factor in what kinds of data can be obtained as source material. As Alex Galloway (2007, 566) notes, the use of the Internet as "one giant database, an input stream that may be spidered, scanned, and parsed" constitutes a distinctive methodology that is notable for its awareness of a principal quality of computational media—"the fundamental mutability of data." The creative use of the network as a medium of storage, access, and exchange is thus characteristic of computational information design and suggests yet another facet of what it means to do design with computation.

Taken together, procedurality, transcoding, and the network as a medium of storage, access, and exchange are primary qualities of computational media. To do computational information design requires an understanding of and agility with these qualities. Regardless of whether one uses Ben Fry's seven stages or another model of information design, these qualities of procedurality, transcoding, and the network as a medium of storage, access, and exchange have affected information design by providing new means of acquiring, organizing, transforming, and presenting data. For the most part, the uses and purposes of information design remain the same—providing structure and form to data to impart greater comprehension in communications across formats as different as news media and scientific publications. Some examples of information design, however, use these qualities of computation for political ends, demonstrating the possibilities of an agonistic information design.

Revealing Hegemony

The visualization *State-Machine: Agency* demonstrates how designers and artists can combine computation with the practices and forms of information design to produce political expressions by rendering data in new ways. Such artifacts and systems of computational information design can be agonistic because they expose and document patterns of association in the construction, maintenance, and exertion of influence in contemporary society. More specifically, they are engaged in an agonistic tactic that I term *revealing hegemony*, which draws on the discussions of hegemony throughout the discourse of agonistic pluralism but specifically in the works of political theorists Ernesto Laclau and Chantal Mouffe (Laclau and Mouffe 2001; Mouffe 2000a, 2000b, 2005a, 2005b).

The concept of hegemony is central to agonistic pluralism. In *Hegemony and Socialist Strategy*, Laclau and Mouffe (2001) build on the work of Antonio Gramsci (1971) to reassess and redefine the concept of hegemony. For Gramsci, hegemony was a class struggle. In his attempt to understand why communist revolutions had not taken hold more broadly, Gramsci theorized that the ideas of the dominant group, in this case the ruling capitalist class, were absorbed by the workers through social structures, including schools and popular media. These social structures lured workers into supporting the capitalist system and ignoring thoughts or actions that would, from Gramsci's perspective, serve them better. Thus, the term *hegemony* broadly describes the way one group develops dominance over another group not by force but by obtaining implicit consent from the subordinate group through social manipulation.

Laclau and Mouffe's work makes two primary contributions to the concept of hegemony. First, they reject the essentialism of Marxist thought present in Gramsci, which defines hegemony from a position of class. Second, they redefine hegemony as an open and flexible discursive strategy. That is, hegemonic practices freely and dynamically bring together histories, ideas, and intentions from a diversity of perspectives into issue-oriented ideologies.

From the perspective of a theory of an agonistic pluralism, hegemony is not a fixed and final state or an effort with a unidirectional vector (from powerful to subordinate). Rather, hegemony is a flexible and vigorous web of related factors, actions, intentions, and objects that are in constant flux, under pressure from and exerting pressure on a multiplicity of positions. This view of hegemony shifts the agonistic effort away from striving to overcome hegemony and toward participating in an ongoing process of exposing and documenting current hegemonic practices so they can be examined and questioned.

Identifying and making hegemonic forces and their means known is vital to the discourses of agonistic pluralism because it helps people discover and label sites and themes of contention in the political landscape. Likewise, the tactic of revealing hegemony through design provides the basis for further agonistic efforts through design or by other means. *Revealing hegemony is a tactic of exposing and documenting the forces of influence in society and the means by which social manipulation occurs.* Designers and artists can use forms of computational information design to represent and perform the associations and flow of resources between people, organizations, and practices, which structure and exert force in contemporary society. To do so, they use the principal qualities of computation as a medium to produce artifacts and systems that uniquely express and respond to the varied and dynamic structures of contemporary hegemony. Two examples that are examined here are social network visualizations and software extensions.

Social Network Visualization: Charting the Associations of Hegemony

Social network analysis is a research method for investigating the relationships between actors within a social context. This method of analysis can be used to examine any network of social relationships and exchanges. It can investigate patterns of gossip within a high school or citation patterns across an academic discipline. What is central to social network analysis is not the content of the relations and exchanges but rather the process of identifying and representing the structure of those relations and exchanges.

Within social network analysis, the analytic activity and visual form are reciprocal. The method produces a visualization that orders and depicts the relations between chosen social actors in the chosen context, and it does so in a manner that allows those relationships and various permutations of them to be documented and explored. Social network visualizations are produced by codifying the relationship between actors in the system, marking their connections, and often assigning weights (representative numeric values) to those relationships to denote the strength of the connections between any two or more actors. Various algorithms and software packages can be used to process these relations and produce a depiction of actors arranged in space such that their spatial distribution from other actors and at times the visual quality of the connection to other actors is meaningful to the relationship status under investigation.[9] From the resulting image, a researcher might be able to make claims about strong or weak ties among members, expose connections among individuals or groups that might otherwise have gone unnoticed, or distinguish individuals or groups for specific interventions to disrupt or bolster the strength of the network. Social network analysis and the commensurate visualizations thus lend themselves to the tactic of revealing hegemony because it provides a method and form attuned to charting the associations and flow of resources among people, organizations, and practices.

They Rule (2001, 2004, 2011)[10] and *Exxon Secrets* (2004)[11] are two projects by Josh On that employ social network analysis and visualization techniques to represent the structures and patterns of influence across corporations and other institutions. *They Rule* (figure 2.2) is an online interactive social network visualization that allows users to explore a dataset containing the names of Fortune 100 companies and their board members. Specifically, with this visualization, users are able to produce images that depict the cross-affiliations among board members of Fortune 100 companies. A user may begin constructing the visualization by either company, institution, or person. If a company or institution is selected it appears on the screen and the user can select to view the board members of that institution or company, which appear arrayed around the image of a board table. Clicking on board members then draws lines connecting to all of the other boards they sit on. The user can continue the exploration process with each member. As the image of the network is constructed, the appearance of the board member is manipulated: the more boards that board members sit on, the fatter they grow. Users can also choose to begin with a single person, and then explore the connections between that person and the boards they sit on. In addition to self-directed exploration of the dataset, users can automate the process of finding connections between companies or institutions. By selecting Find Connection from the menu, a user can

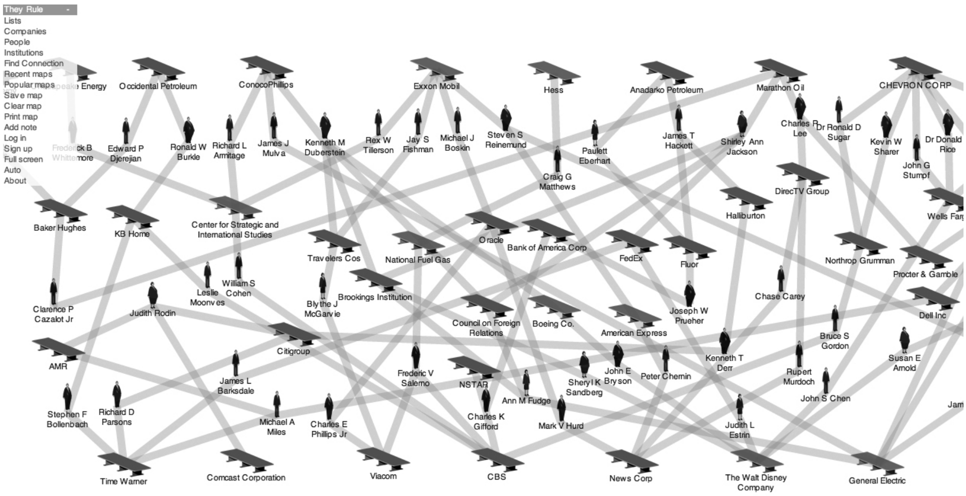

Figure 2.2
Josh On, *They Rule* (2001, 2004, 2011), http://www.theyrule.net. Map created by user ssackz, on the 2011 version of *They Rule*. This map depicts the relationship between the oil and media industries and is titled *Oily Media*.

select any two companies or institutions to be automatically searched, and the software produces a visualization that shows how those two companies are connected via their board members. As yet another mode of interaction, users can save maps they have created, thereby contributing to an ongoing archive of documented structures and patterns of connection between corporate boards.

The social network visualization *Exxon Secrets* (figure 2.3) is built from the same software code base as *They Rule* but is focused on a different domain. Sponsored by the environmental advocacy organization Greenpeace (USA), *Exxon Secrets* is a social network application that exposes and documents the influence of the petroleum company Exxon-Mobil in shaping climate-change debate, regulation, and legislation. The visualizations chart the funding that the Exxon Mobil Foundation provides to researchers, lobbyists, and organizations that are climate-change skeptics and who question the reality or significance of climate change. The implication of this visualization is that the Exxon Mobil Foundation uses networks of funding to influence the climate-change debate by supporting research and public relations efforts that counter claims that climate change is a detrimental phenomenon that is in part caused or exacerbated by fossil fuel production and use. The purpose of the *Exxon Secrets*

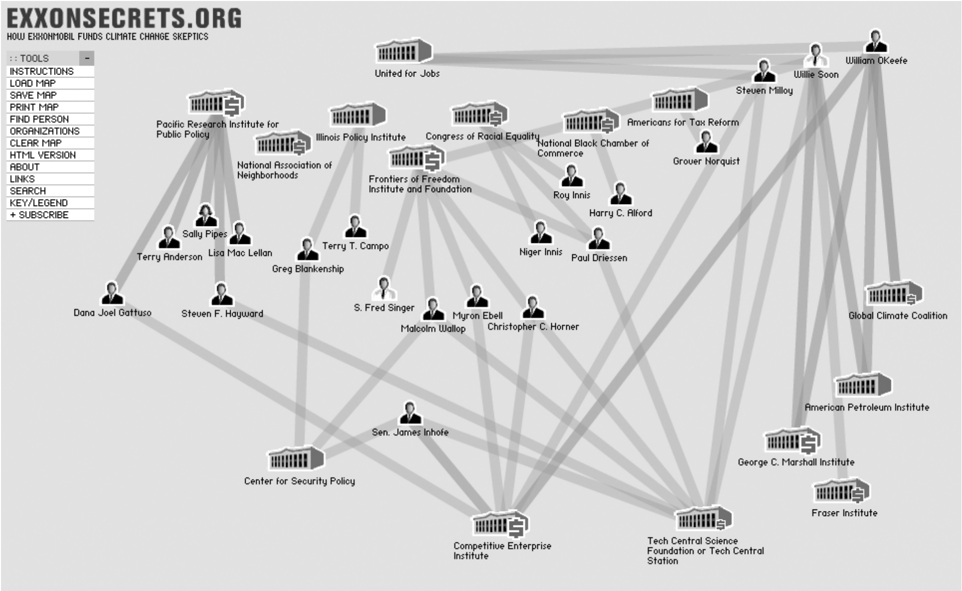

Figure 2.3
Josh On, *Exxon Secrets* (2004), http://www.exxonsecrets.org/maps.php

visualization, then, is to present the complex lattices of financial relations that shape this discourse and action.

The actors in *Exxon Secrets* are scientists, spokespeople, and organizations and are categorized by funding amounts. Instead of *They Rule*'s icons of boardrooms and businessmen and -women, *Exxon Secrets* uses icons of government buildings (overlaid with dollar signs that change in size) and icons of heads (differentiated by gender and role). As with *They Rule,* users can construct network representations either by organization or individual. The user then can show all of the organizations that an individual is connected to or all of the individuals connected to an organization. For each individual or organization, a user can view a panel of detailed information, including background information, notable quotes and deeds, and a timeline of funding received from the Exxon Mobil Foundation. In addition, as with *They Rule*, *Exxon Secrets* includes a set of premade visualizations.

As examples of agonistic computational information design, both *They Rule* and *Exxon Secrets* chart associations among institutions, individuals, and issues, with the implication that these relations form a structure through which influence is exerted. In *Exxon Secrets*, these relations are complex and involve myriad actors and events that intersect with the subject of climate change. The inclusion of events and organizational details in *Exxon Secrets* is a significant addition to the functionality of *They Rule*. It extends the capacity of the software to reveal hegemony because it allows users to discover patterns of action as they develop and change over time in response to specific issues. By showing how actors, resources, and interests align in variable configurations that are not reducible to simplistic or obvious patterns, the visualization expresses the dynamic and contingent nature of hegemony. For example, with *Exxon Secrets*, users can view and compare the networks of associations formed around "Climate Stewardship Act Attack" with those of the "Global Warming Legislation Attack." With these two visualizations, the arrangements of individuals and institutions across the events can be studied to develop an understanding of the political landscape of climate change generally and around each of these events specifically.

By providing views into the actors and forces involved in a particular issue, *Exxon Secrets* works to represent hegemony as a conglomeration of ideas and intentions from a diversity of sources. Hegemony, so depicted, is not a structure or condition based on class distinction. Instead, it is a condition of associations and attachments to an issue—in this case, to work against climate-change research and associated legislation. This notion of associations and attachments, more so than dividing lines along the class, status, or even political party affiliation, characterizes the contemporary understanding of hegemony. Because of their formal qualities, social network visualizations are particularly suited to provide representations of these connections and orderings of resources to issues. Moreover, by providing simple capacities for interactivity with the visualization, these projects also suggest the potential to use computational media to produce ever more dynamic views into the construction and exertion of hegemony in contemporary society.

Nearly a decade after the first instantiation of *They Rule,* social network analysis and visualizations continue to be used by artists and designers to examine hegemony. However, with changes in the context and capabilities of computational media have come changes in the processes and products of computational information design and its agonistic variants. Creating *They Rule* and *Exxon Secrets* required that On search, locate, and organize the data that underlie the representations, but today such data are more readily available, formatted, and able to be accessed online. An increasing number of tools also automate or at least proceduralize the visual formatting and display of data. From this confluence of factors comes the mashup—a distinctive practice and form with significance to computational information design.

The mashup as form derives from music and is akin to the remix: it is a combinatorial form made from the blending of preexisting parts.[12] In computational information design, a mashup is data (of any kind) that come from two or more sources and are brought together to produce a new form or functionality. In most domains of computational media, the term *mashup* refers to a Web-based software application that draws on data and presentation layers from two or more digital applications or services to produce a third, distinctive, digital application or service. Mashups often rely on application programming interfaces (APIs), which are sets of access points and exchange techniques that allow a programmer to write software that accesses data from one application or service and passes it to another. The central activity of creating a mashup is using code to suture together data into a new form, and it is made possible by the malleability and interoperability of digital data—the qualities of computational media that enable transcoding.

The visualization *State-Machine: Agency*, discussed at the start of this chapter, is one example of computational information design via the mashup: it draws from a series of online databases and other resources to acquire the data that it combines and then expresses. Another example is Skye Bender-deMoll and Greg Michalec's the *Unfluence* project (2007)[13] (figure 2.4). Like *They Rule* and *Exxon Secrets,* the *Unfluence* project uses a social network visualization and computational information design to reveal patterns of influence across institutions, organizations, and individuals. Like *State-Machine: Agency,* those networks and forces of influence concern campaign contributions. In fact, both *State-Machine: Agency* and *Unfluence* were winners in the Sunlight Foundation's[14] 2007 mashup contest that sought innovative examples of the use of computational media to promote government transparency. What distinguishes the *Unfluence* project from *State-Machine: Agency*, *They Rule*, and *Exxon Secrets* is the extent to which it draws from other sources and the capabilities it provides for exploration. As an artifact of computational information design, the *Unfluence* project exemplifies a procedural approach to information design and highlights issues regarding the role of the image in evoking the political.

Arriving at the Web page for the *Unfluence* project, a user specifies the state, year, government office (governor, state senate, state house, state assembly, state supreme court), category of people from whom the contributions come (political action constituents and lobbyists, such as accountants, health professionals, conservative Christians, or pro-life activists),

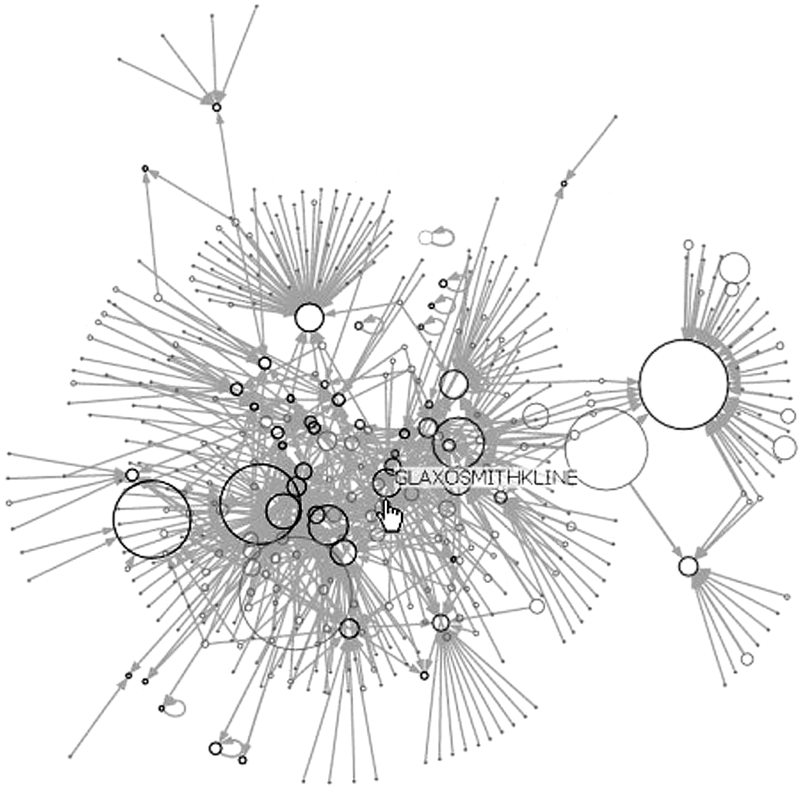

Figure 2.4
Skye Bender-deMoll and Greg Michalec, *Unfluence* (2007), http://unfluence.primate.net

and a dollar amount (such as greater than $500, greater than $1,000, or greater than $10,000). The user then clicks on Generate Graph, and a social network visualization is produced that represents all the contributions to all the candidates within the specified election and contribution size. Users can then explore the visualization. Circles representing donors are green, their size is relative to donors' total contributions. Arrows emanating from the donor circles mark their connections with the candidates. The circles of candidates are red or blue, depending on their political party affiliation, and their size represents the total amount of money received from all contributors. Pointing at any circle (that is, any candidate or donor) displays the name of that individual or organization. Clicking on the circle opens an auxiliary window that searches for and retrieves information (if present) regarding the individual's voting record or past contributions.

Bender-deMoll and Michalec (2007) describe the process of generating the *Unfluence* visualization by detailing what this design process involves:

> A query is generated from your search settings and sent to National Institute on Money in State Politics' API which looks in their databases and returns a list of matching candidates as an xml file. For each candidate we get a list of the top contributors, and discard any with contributions below the value threshold you set. This donor-recipient information is formatted into a network and passed to a program called GraphViz that computes positions for the nodes and draws it (with help from ImageMagick). The image is passed back to you. When you click on a node, we send queries to NIMSP and Project VoteSmart to check if there is information available (this requires some hacked scraping and matching code) for that candidate, and include the links in the info bubble. The visual effects are provided by script.aculo.us.

This description conveys the ways in which computational information design is increasingly a practice of procedurally acquiring, referencing, and combining data. The artifact that results from this process—the image itself—does not exist a priori and is not set in its final appearance by the designer. Instead, the image emerges from and is procedurally expressive of the data and the code written to render that data. That is, the form that the designer provides is not the final form of the image but rather the rules from which to construct the image. This mode of producing representations is characteristic of computational artifacts and systems and marks a point of distinction from other media forms. As Bogost (2007, 4) describes, "To write procedurally, one authors code that enforce rules to generate some kind of representation, rather than authoring the representation itself." In the case of *Unfluence*, even the formal qualities of the image of the network—its so-called look and feel—are the direct effect of the software libraries used.

Varieties of Political Expression

All of these computational information design projects engage with content that is political in nature, and all of them seek to show patterns of financial influence in and across elections, governance, and corporate organizations. These visualizations are agonistic because they provide access to hegemonic conditions, making them seen and knowable. But even within this tactic there are varieties of political expression that can be differentiated.

A social network visualization is not inherently political, but its form *lends itself* to visualizing hegemony because it can represent the relations between a heterogeneous array of entities. More generally, a visualization is not political or more political just because it employs computation. Although the design of *Unfluence* leverages the principal qualities of computational media more thoroughly than *They Rule* or *Exxon Secrets* does, that does not necessarily make it more agonistic. Indeed, its provocation is arguably less than that of *They Rule* or *Exxon Secrets*. When users view the image of a network in *Unfluence*, there is nothing in the form of the representation that is explicitly contestational. In contrast, consider the images of the networks in *They Rule* and *Exxon Secrets*, where the networks are presented as visually distinctive images that are politicized by means of visual design. The form is manipulated to communicate decidedly politicized perspectives. In the design of *They Rule* and *Exxon Secrets*, On provides visual anchors via icons that give meaning and assign identities to the actors within the network, visually casting those actors in roles that evoke negative connotations and that are unquestionably political. The board members in *They Rule* grow fatter as they join more and more boards, conveying associations to excessive consumption, and the institutions in *Exxon Secrets* are overlaid with larger and larger dollar signs as the contributions they receive from Exxon increase. These anchoring icons in *They Rule* and *Exxon Secrets* leave the user with few doubts about the political positions that are expressed through these visualizations.

Using visualizations to express political positions introduces a bias into the form, thereby distancing these visualizations from their social scientific counterparts that strive to report without prejudice. From an agonistic perspective, the bias in these expressions is appropriate, not problematic. Centrality, or neutrality, is impossible in agonistic pluralism because the broad and divisive differences of positions are considered to be constitutive of the political condition (Mouffe 2005b). *Bias is required to do the work of agonism*. A visualization that is agonistic cannot just present the facts. An artifact of information design is made agonistic by the extent to which it identifies and represents contestable positions or practices. Given that the tactic of revealing hegemony is meant both to document hegemonic conditions and also to rouse and shape future arguments and action, artifacts and systems engaged in this endeavor combine political content with unabashedly biased visual representations that work vigorously as provocations.

Unfluence and the project that began this chapter, *State Machine: Agency*, are both concept works. They are responses to a solicitation for projects from the Sunlight Foundation, and they show how social network visualizations can provide an awareness of the entanglement of money and politics and of the structures and patterns of influence. These projects do important work as demonstrations, but it is worth noting that *Exxon Secrets* functions as something more. Of the visualization projects discussed so far, *Exxon Secrets* connects most strongly with common notions about the practice of design. With *Exxon Secrets*, an artifact has been constructed for and situated within the context of a project beyond itself. *Exxon Secrets* provides an example of how agonistic information design can fit within a broader political project as a component of communications and advocacy efforts. It shows how social network visualizations can reinforce an adversarial stance between two sides of a political conflict—the climate-change debate. The visualization is called on to do a particular job—to provide evidence for an argument that corporate forces are aligned against climate-change research and legislation. Situated within the Greenpeace organization, the visualization does double duty—as a resource for those already engaged with the issue and wanting to understand more about the influence of these networks and also as an incitement to those unfamiliar with the issue, drawing them in and providing pathways to gather more information or take action through Greenpeace (figure 2.5).

Exxon Secrets also provides an example of how procedurality can support a practice of agonistic information design by enabling the replication of politicized forms across issues. As with *Unfluence* and *State-Machine: Agency*, the project *They Rule* can be considered to be a demonstration because it provides a compelling example of the potential of computational media to evoke the political. This potential is most fully realized when the visualization is used for multiple political provocations. That is, the capacity of procedurality is most acutely evident when On reuses the code base from *They Rule* for *Exxon Secrets,* shifting with relative ease from charting the associations among the boards of Fortune 100 companies to charting the associations between individual corporations and their relations with institutions around an issue. This requires the accessing and parsing of new datasets and the designing of new icons, but the core structure and capabilities remain, carried in the code, and can be applied over and again. This capacity for the procedural transfer and thus repeated instantiation of tactic and form from one contested topic to another is one of the outstanding qualities of computational information design.

Extensions as Interventions

Visualizations are a common and compelling form, but they are not the only form of computational information design. Other forms and practices of computational information design rely less heavily on the image in the

Figure 2.5
Josh On, *Exxon Secrets,* shown as embedded in the Greenpeace USA Web site (2011), http://www.greenpeace.org/usa/en/campaigns/global-warming-and-energy/exxon-secrets

familiar sense that visualizations do and thus demonstrate other ways information design might be directed toward revealing hegemony. One example is the design of extensions for Web browser software. Extensions are supplemental software applications that can be installed to run in conjunction with another core application program, augmenting that application with whatever functionality is made possible by the extension. For example, there are extensions that block online advertisements, that add artistic flourishes to the software's visual appearance, and that streamline productivity by assisting in online information management.[15] Most of the time, extensions are used for simple things, such as adding more rows to a Web browser bookmarks bar, but there are also inventive uses of extensions for decidedly political expressions.

Extensions are similar to mashups in that they leverage the capacity for transcoding: they access data from one application or service and pass it to another to produce a new functionality. The primary difference between mashups and extensions is that the mashup constitutes a new application and the extension is embedded in another application. This embeddedness presents an opportunity for design and a variation on the tactic of revealing hegemony. Whereas visualizations are fundamentally about depicting in a graphical manner, extensions can be used agonistically as interventions that both expose and contextualize hegemony. With extensions operating as interventions inside existing applications, the tactic of revealing hegemony is extended and amplified by the ability to *reveal in place*, connecting the exposure of hegemony together with the conditions through which hegemony occurs.

Designed and implemented by Nicholas A. Knouf (2009), the *MAICgregator*[16] is an intervention by way of computational information design (figure 2.6). Short for Military Academic Industrial Complex Aggregator, *MAICgregator* is a Firefox Web browser extension that exposes and contextualizes the relationships between academic research and military funding "to counter the hegemony of the present-day University" (Knouf 2009). The added functionality of the *MAICgregator* that is provided by the extension is not a productivity enhancement or ornament. The *MAICgregator*

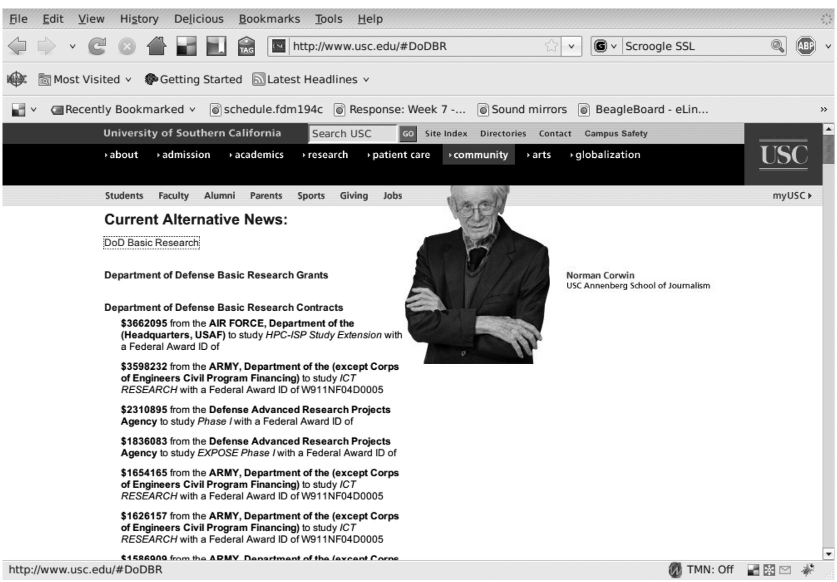

Figure 2.6
Nicholas A. Knouf, *MAICgregator* (2009), http://maicgregator.org showing the Department of Defense funding for the University of Southern California in 2009

intervenes in the Web pages of universities located in the United States and provides information about military funding that they receive, with links to external files on the Internet that document that funding and the associations between academic research and defense initiatives. The *MAICgregator* goes beyond depicting the networks of hegemonic forces and provides a reflexive investigation of hegemony. It reveals the ways that military funding, research, public relations, and news media mix together in the contemporary university and contextualizes this revealing within the Web site of the institution under examination. In contrast to visualizations such as *Exxon Secrets*, which provide a view of hegemony from the outside looking at an issue, the *MAIGgregator* provides a view of hegemony from the inside.

After installing and activating the *MAICgregator* extension, a user can visit a university's Web page, and if the school receives military funding, then headlines, short text descriptions, and links to details regarding that funding are inserted into the page. Depending on the settings of the extension, this information may surreptitiously replace the underlying, original content. For example, the news section on a university's home page might be replaced with a new headline "Current Alternative News," which shows links to public relations announcements for military research associated with that institution. By adjusting the settings for the extension, the user can allow images of university trustees to be inserted into the layout of the page, replacing the existing images. In some cases, the replacement of text and image is nearly seamless, with the military funding data and images of trustees being integrated into the existing structure of the Web page to appear as if it was the original and intended content of the page. In other cases, the integration of the auxiliary data is not smooth, resulting in layouts that range from the slightly awkward to the chaotic. In either case, the experience of navigating and consuming the Web with the *MAICgregator* can be disconcerting, as images are replaced with new images that do not quite fit either in scale or style, and links are inserted into pages that unexpectedly lead users off the site to press releases for often obscure research projects.

The *MAICgregator* is yet another example of the procedural rendering of data in computational information design. Here, again, the activity of design is not the authoring of a specific and predetermined representation but rather the authoring of software, composed of rules that, when executed, produce a representation. In the case of *MAICgregator*, the representation attempts to be relatively unexceptional. It is designed to be integrated into a given form—the existing format of an academic Web site. This is not a trivial design task. To achieve this culling and integration of data requires an understanding of the location and structure of data across the Internet. Knouf and his collaborators detail this process in their project documentation.[17] In brief, the *MAICgregator* searches the Internet for sources of information concerning military funding of academic research and focuses on United States Department of Defense funding. Sources of data include the USAspending.gov database for grants and contracts; the DOD Small Business Technology Transfer program database; the PR Newswire database; the Foundation Center 990 Finder database to locate trustee names; a Google News search for relevant news stories; and a Google image search to locate images of trustees. The design challenge goes beyond locating and retrieving data, however, because the data also must be parsed and correlated. That is, for the data to be transformed into information, they must be identified by type and associated with specific academic institutions. Finally, the design of a given page of an academic institution must be deconstructed so that the appropriate information can be integrated back into that page in the correct places. Integrating all of these processes together makes for the functionality and experience of the *MAICgregator*.

Like the *MAICgregator*, the *Oil Standard* (Mandiberg 2006)[18] is a Firefox extension that integrates auxiliary information into Web pages to document associations and effects within hegemonic social conditions. As with the *MAICgregator*, the issue of concern is grounded in economics, but here there is a shift of focus and content. Created in 2006 by Michael Mandiberg, the *Oil Standard* extension replaces or augments the monetary amounts in any given Web page with their equivalent cost in barrels of crude oil. The extension's name is a play on terms that refers to the gold standard, which grounded the value of money in gold as an objective reference point. This project makes clear that the standard is no longer gold but oil and that the standard is not as objective or at least not as fixed as gold was.

Rather than simply provide the daily cost of oil, a number that is readily available elsewhere, *Oil Standard* transforms the data so that they are more understandable, grounded, and meaningful. Depending on the preferences set by users, either the standard price in U.S. dollars is replaced entirely, or the price in oil is placed next to the standard price in U.S. dollars on all Web pages. This includes the cost of items for purchase on commerce sites and any monetary amount listed on a page. So when the line "$260 billion" appears in the text of a news story regarding national debt, next to it in parentheses appears the conversion of that amount into numbers of barrels of crude oil. Likewise, when a user purchases a book or mp3 player, the cost of the item is translated into numbers of barrels of crude oil.

Oil Standard engages in another variation on the tactic of revealing hegemony, which extends the act of documenting and can be characterized as an endeavor of *translation*, which is concerned with the invention and expression of equivalencies between constitutive elements of hegemony. With *Oil Standard*, this process of translation begins with the construction of equivalencies between two constitutive elements—oil and money. To this is added a third—objects of consumption (such as the items on Web pages with oil prices associated with them) (figure 2.7). These objects of consumption ground and express the relation between oil and money in a way that enables understanding. In *Oil Standard*, the objects of consumption—whether a paperback book or an mp3 player—operate as the translating elements as they are transformed into equivalencies with oil. Compared to the cost of oil, the perceived value of everyday objects remains relatively more constant over the short term. We have a general idea of how expensive and valuable an mp3 player and a paperback book

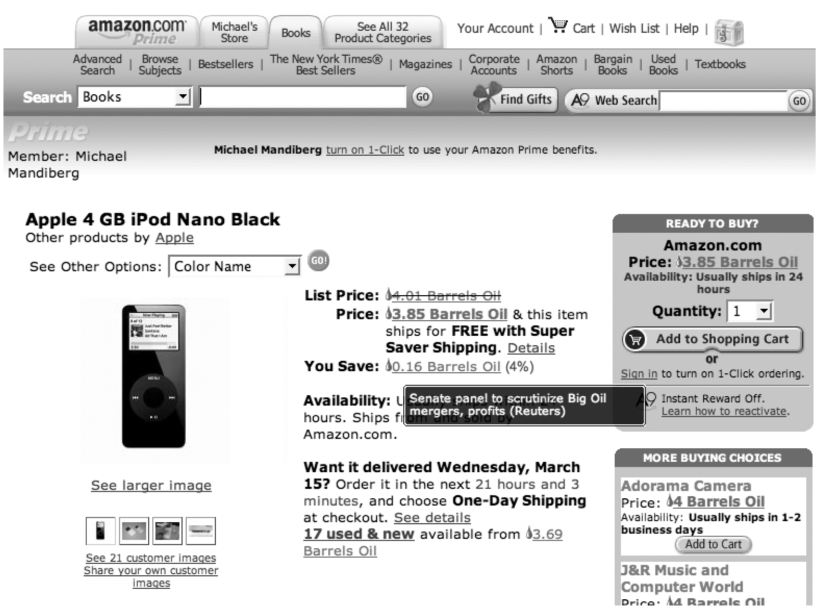

Figure 2.7
Michael Mandiberg, *Oil Standard* (2006), http://www.turbulence.org/Works/oilstandard

are. By the conversion of money into oil and then consequently the conversion of the cost of objects into oil, users are provided with a grounding of the value of oil that is experientially accessible and understandable and reinforces the hegemony of oil, into which all things of value can be and are converted.

As examples of agonistic information design, the *MAICgregator* and *Oil Standard* extensions weave together technical capacities and use. On a technical register, they operate by interceding in and augmenting software, adding new functionality and purpose. This intervention occurs at the level of data and code and thus draws attention to the possibilities of a technical, specifically computational, form of expression with political intent and affect. Although extensions are not inherently political, this particular form of political expression would not be possible outside of computational media. It depends wholly on the combined qualities of procedurality, transcoding, and the network as a medium of storage, access, and exchange.

These extensions as interventions are a form of computational information design that is particularly appropriate to expressing hegemony, and they demonstrate the ways in which the qualities of computation can be leveraged to evoke the political. As Laclau and Mouffe redefine hegemony (2001), it is a dynamic and contingent combination of histories, ideas, and intentions from a diversity of perspectives: hegemony is as a constantly changing arrangement of forces and effects. Because of the technical capacities of the extension to aggregate and integrate data in near real-time, it is uniquely capable of documenting and expressing these dynamic and contingent conditions. For example, the effects of *MAICgreator* vary according to research, funding, and news cycles. As university projects develop, receive funding, get promoted through institutional and governmental public relations departments, and are picked up by the news media, the content of the information integrated into a given university's Web page changes. Through the procedural lens of the *MAICgregator*, a given university Web page in September 2010 might appear markedly different than the page would have appeared in September 2009, due to the changing status of funding associations. A similar situation is present with the *Oil Standard*. As the price of crude oil fluctuates, the translated cost of an iPod, a copy of *Pride and Prejudice*, or any other item that appears with a price in dollars on a Web page will also vary. Even though the cost in dollar amounts has not usually changed for these items from one day to the next, their value in crude oil has. In both cases, the technical capacities of the extension produce expressions that reflect the variable constitution of forces and effects that characterize a contemporary understanding of hegemony. In both cases, computational information design is used to express the persistent interleaving of influence throughout our everyday activities and familiar social institutions.

Extensions as interventions also operate along a register of use. Like other forms of interactive computational media, such as video games, extensions are experienced only when they are run,[19] in this case within the host software of a Web browser. One cannot simply launch an extension and have it return data or information without using the software that it extends. Put another way, with extensions, actual use is required to evoke the political. This is significant because in these cases, the activity of revealing hegemony reflects user actions and the contexts in which users find or place themselves—for example, browsing academic Web sites or shopping online. Because the effects of these extensions are shaped by one's own interests, choices, and actions, the process of revealing hegemony becomes personalized, contextualized to one's self as a consumer of information and goods. Through notions of *revealing in place* and *translation*, which are made possible by weaving together technical capacities and use, the idea of hegemony shifts from a generic notion of external forces —a vague specter—to an experience of hegemony in which users themselves are present as actors. The notion of use adds another dimension of note to agonistic computational information design, and it provides another manner of distinguishing these works—by comparing expressions that represent and those that perform.

Representing and Performing

All of the projects presented in this chapter contribute to a common endeavor of revealing hegemony, but there is a key difference in *how* they do so. To varying extents, they all leverage qualities of the medium of computation to produce representations, but some go further to produce systems that perform the very conditions of hegemony that they strive to reveal. In doing so, they constitute a mode of adversarial design that is unique to computational media and further demonstrates the ways in which computational information design can do the work of agonism.

Projects such as *They Rule*, *Exxon Secrets*, and *Unfluence* produce representations of hegemony: they graphically depict networks of force, influence, and the means of social manipulation. These representations provide illustrations that document the various actors involved in particular hegemonic conditions and allow users to explore variations across those collections of actors. For example, with the interface elements of these designs (such as menus and check boxes), a user can select different corporations, individuals, or events by which to structure the visualizations. Because the resultant images are procedurally generated, it is possible to produce extensive series and variations of representations with relative ease. With this ability to select among actors and thereby produce different views, the representations begin to provide a perspective on hegemony that aligns with Laclau and Mouffe's notion of hegemony as manifold and multifaceted. Through these representations, users can move beyond understanding hegemony as simply a single-point exertion of force. They are given a view into the constitution of hegemony as a flexible conglomeration of individuals, organizations, ideologies, and actions. They also can analyze the extent to which the particular visual forms of a representation are political—that is, the extent to which they explicitly communicate a contestable position.

Project such as *State-Machine: Agency*, *MAICgregator*, and *Oil Standard* extend the means of graphical depiction and operate in a distinct manner. These projects produce representations but they also perform the hegemonic conditions that they reveal. The hegemonic conditions are procedurally enacted as a user interacts with or makes use of the software.

Consider again *State-Machine: Agency*. The political stance that is advanced through the visualization is performed through the expressive qualities of the visualization. The software that structures the visualization procedurally enforces relations between datasets, visually formalizing and kinetically expressing a relationship between politicians and money. So with the data and the algorithmic structuring of the work and the affordances of interactivity, *State-Machine: Agency* performs this condition of influence in contemporary politics. When interacting with the visualization, a senator's position on the screen is defined by his or her relation to the selected funding sources, and it is impossible to separate a senator from these funding sources. Although a user can click and drag a senator away from his or her funding source momentarily, the circle bounces back into place as soon as the user releases the button, thus procedurally performing a claim about the politician's binding relations to campaign contributions.

Even more than *State-Machine: Agency,* the projects *MAICgregator* and *Oil Standard* perform the hegemonic conditions they seek to reveal by way of their technical format. By integrating with the structure of a university Web site or the activity of consumption—that is, by integrating the project with another context and action—they perform the pervasiveness that is characteristic of hegemony. For example, when a user casually shops online with *Oil Standard* installed, she ubiquitously encounters the value of oil. As long as the extension is running, there is no escape from the collapse of all values into the currency of oil, thus performing the notion of the influence of oil as being all encompassing. The *MAICgregator* also imbues users with a sense of the sweeping entanglement of academic research and military funding. Both projects also offer an aesthetic strategy of seamlessness that reinforces the pervasiveness of hegemony.[20] As the integration of the information concerning defense funding or oil prices is incorporated—by way of transcoding—into the experience of surfing the Web, it enacts the way in which hegemony operates by efficiently interweaving ideology and influence into social structures and everyday activities.

MAICgregator and *Oil Standard* thus provide demonstrations of how information design and computation might be brought together to construct new forms of adversarial political expression. In these projects, the conditions and constructs of hegemony are literally codified in the design. And moreover, with *MAICgregator* and *Oil Standard*, hegemony is brought to the fore *in situ*. It is encountered in a mediated form that calls attention to itself by both its visual presence and its pervasiveness. Such examples of agonistic information design operate in a manner similar to Bogost's notion of a procedural rhetoric in video games, which "represent how real and imagined systems work . . . [and] invite players to interact with those systems and form judgments about them" (Bogost 2007, vii). Like the video games that Bogost describes, these examples of agonistic computational information design invite users to experience the conditions and constructs of hegemony, develop an understanding of hegemony, and perhaps form judgments about those conditions.

Summary

In his essay "Critical Visualization," Peter Hall (2008, 128) calls on readers to consider visualization as "a creative process concerned with not just the finished artifact but the framing, gathering, connecting and arraying of data" to "imagine it as a critical practice: sizing up and reformulating a terrain of knowledge as well as experimenting with new and alternative forms." This chapter presents examples of such a practice of critical visualization and critical information design that do the political work of agonism. As Mouffe (2005, 25) states, "Mobilization requires politicization, but politicizing cannot exist without the production of a conflictual representation of the world, with opposed claims, with which people can identify, thereby allowing for passions to be mobilized within the spectrum of the democratic process." One of the tasks of agonistic information design is to provide those conflictual representations of the world. The examples in this chapter depict claims concerning the structure and exertion of power and influence in contested matters such as campaign finance, corporate leadership, policy and the environment, military research, and oil.

If agonism is taken to be an ongoing endeavor of politicizing issues, then revealing hegemony is perhaps the most basic tactic of this endeavor. As a tactic, it works to make conflictual positions better known and better available for contest. When analyzing adversarial design, one question to ask is, How and to what extent does a given artifact or system of computational information design engage in the tactic of revealing hegemony?

Answering that question requires investigating the ways that the forms of information design are combined with the principal qualities of computation to render artifacts that are decidedly political. As discussed, the artifacts and systems of computational information design are particularly suited to revealing hegemony because the principal qualities of computation can be used to express the dynamic and associative qualities of influence and social manipulation. Hegemony, as is discussed here and within theories of agonistic pluralism, is not reducible to class distinctions or unidirectional relations from the so-called powerful to the subjugated. Rather, this reconsidered hegemony extends in all directions. Just as the condition of hegemony is heterarchical, organized through associations to issues, so too should be the efforts to expose and document, represent, and perform hegemony. Computation as a medium provides distinctive affordances for the political expression of hegemony because of its capacity to render large amounts of ever-changing data from many sources and in many formats. In addition, computation as a medium provides users with the capacity to exert choice in the ordering of that data. Through basic interactivity, users can explore and construct displays of one condition or another, producing representations and performances of hegemony that are reflective of the interests, desires, and in some cases, the actions of the users themselves.

One challenge with the tactic of revealing hegemony is to move beyond simplistic forms of demystification, as if the hegemonic condition was unknown. Too often there is an assumption that simply showing or stating something is an important political act. In some cases, this may be true, but it is important to move beyond just raising a general awareness of a situation. Critical faculties are not needed to discern that special-interest groups and political action committees contribute to election campaigns, that corporations fund research to advocate for policy in their best interest, that universities are entwined with military and intelligence agendas, and that reliance on oil affects all modes and manners of consumption. But the examples in this chapter demonstrate possibilities beyond simply exclaiming, “Hegemony exists!” The examples in this chapter suggest how computational information design might work to delve into and communicate the particularities of hegemonic conditions in novel ways—vividly recording and providing evidence of the associations and flow of resources between people, organizations, and issues, which goes beyond simplistic declarations of the already known.